Minimum Embedding Inference

A Unified Pipeline for LLM-Grounded Bayesian Coherence Scoring and Cross-Domain Transfer in Belief Networks

Anonymous submission

Abstract

Setting. A rational agent choosing among competing explanations prefers ones that fit existing beliefs with minimum revision [Quine, 1951, Lakatos, 1970]; a rational learner confronted with a novel problem prefers structural patterns it has seen before over structural novelty it has not [Lakatos, 1970, Finn et al., 2017]. We argue these two intuitions are the same object viewed at two scales — one *within* a single belief network (coherence) and one *across* belief networks (transfer) — and we build a single PPL-native pipeline that operationalises both.

System. Minimum Embedding Inference (MEIS) is a four-layer architecture: L1 PyMC belief networks [Salvatier et al., 2016] as the representation, L2 KL-divergence and Fisher-information scoring, L3 a graph-edit-count structural term plus Markov-category morphism matching [Fritz, 2020, Perrone, 2024], L4 an LLM acting as *translator* between natural language and the underlying probabilistic programs. The LLM never produces the final answer; a sampler does. We describe the pipeline end-to-end and report three distinct slices of evidence.

Evidence slice 1 — LLM-grounded inference (§3). On two independent boxing-gym [Li et al., 2025] out-of-domain benchmarks (Dugong growth, Peregrine falcon dynamics), LLM-extracted cross-domain priors routed into PyMC produce a statistically significant improvement at $p < 0.001$ on the scientist-side task. The NL→PyMC handoff is a *bottleneck* — replacing prose with structured JSON cuts MAE by 87% on baseline and 56% on MEIS-full on Peregrines ($p < 0.001$ and $p = 0.003$).

Evidence slice 2 — Coherence ranking (§4). We formalise a *minimum-embedding score* $D(h, \mathcal{B}) = D_{\mathrm{KL}}(P(\theta^* \mid \mathcal{D}) \,\|\, P(\theta^* \mid \mathcal{D}, E_h)) + \lambda \cdot |\Delta_h|$ that decomposes into posterior drift on a target latent and a BIC-style structural cost. We prove that orphan-node hypotheses (new latents d-separated from θ^*) are ranked last whenever $\lambda|\Delta_h| > \max_{h'} D_{\mathrm{KL}}(\theta^* \mid h')$ (a framework property, not an empirical finding). On three constructed benchmarks the framework property holds; a cross-family LLM expert panel of 4 raters (GPT + Gemini) agrees (12/12 rater-benchmark pairs place the orphan last, mean inter-rater $\tau = 0.76$). On a prospective experiment where hypotheses are generated by an LLM rather than the authors, D perfectly respects the LLM-declared coherence tier (in-vocab $\ll$ orphan $\ll$ contradictory, with 212× and 528× separation). Full-BIC model selection disagrees with D on 2/3 benchmarks (Kendall $\tau = 0.00$ and 0.33), demonstrating $D \neq$ BIC in practice.

Evidence slice 3 — Categorical structure transfer (§5). The minimum-embedding intuition extends across networks: two belief networks that share a functional form should share inductive bias. On a law-zoo of 3 equivalence classes × 10 physical domains (exponential decay / saturation / damped oscillation), four independent equivalence-detection layers — PyTensor op-multiset fingerprint, Weisfeiler-Lehman DAG refinement, symbolic Markov-category morphism, and a BSS + Perrone semantic check — all recover the 3-class partition (ARI = 1.00). The agreement is partly *by construction* since the fixtures share a canonical $\mu(\theta, t)$ per class; we report it honestly. On few-shot transfer the signature-gated protocol delivers a mean 89.5% held-out MSE reduction on saturation-class targets and no benefit on exp_decay-class targets (a dynamical asymmetry, not a method failure).

Scope (honest up front). All three evidence slices are on small, partly author-constructed benchmarks; all LLM-involving experiments use at most four models across two families; the GNN embedding we ship as infrastructure does not demonstrably outperform random-init on the current fixtures. What we offer is a single coherent pipeline with three distinct axes of evidence, a reference implementation with 70+ unit tests, and a precise specification of what this formulation of minimum-embedding inference requires and where its current evaluation is weakest.

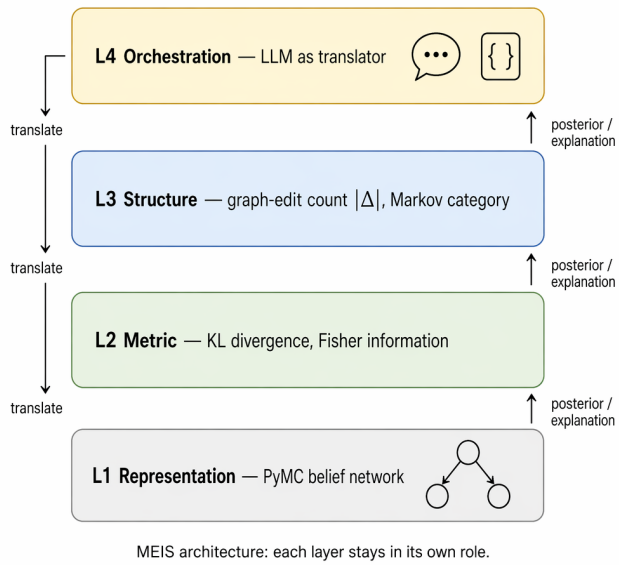

Figure 1: The MEIS four-layer architecture. Information flows down (natural-language problem statement → structured priors → posterior inference) on the left and up (posterior statistics → natural-language explanation) on the right. Each layer stays in its own role: the LLM translates; the sampler infers; the metric layer scores; the structural layer detects equivalence across networks.

1 Introduction

Two philosophically long-standing intuitions about reasoning, one about belief revision and one about skill acquisition, have structurally similar implications. Quine [1951]'s *web of belief* captures the first: a rational agent's beliefs are organised as a graph whose periphery absorbs corrections first, and who should prefer explanations that disrupt the graph minimally. Lakatos [1970]'s research-programme framework captures both: a research programme has a *hard core* defended by a *protective belt*, and a successful new programme preserves the core while modifying the belt; dually, a new problem is tractable in proportion to how much of it reduces to an already-solved one. The Bayesian posterior update $P(\theta \mid \mathcal{D}) \propto P(\mathcal{D} \mid \theta)P(\theta)$ operationalises the first (KL-minimal perturbation of the prior [Williams, 1980]); meta-learning operationalises the second [Finn et al., 2017] via gradient-based initialisation transfer.

Neither operationalisation is complete. The Bayesian posterior update captures the *numerical* half of belief revision (D_{KL}-minimising drift on a target latent) but not the *structural* half — how much a hypothesis requires rewiring the network's graph, which in practice distinguishes coherent explanations from incoherent ones more clearly than any target-drift term does. Meta-learning transfers parameter initialisations but requires the modeller to declare the task-to-task mapping; it cannot *discover* that two tasks are instances of the same abstraction.

This paper argues that these two gaps are the same gap, and that both can be filled by the same type of object: an additive decomposition of a model-selection criterion into a data-fit term and a structural-cost term, evaluated once within a network (for coherence ranking) and once across networks (for transfer gating). We call the full system **Minimum Embedding Inference** (MEIS). The architecture is four layers (Figure 1):

- **L1 Representation:** PyMC probabilistic programs [Salvatier et al., 2016]. Free random variables are latent parameters; observed random variables encode evidence; deterministic nodes encode functional form. Inference is via NUTS [Hoffman and Gelman, 2014].
- **L2 Metric:** KL divergence on posterior distributions (conjugate-closed-form and Monte-Carlo paths); expected Fisher information on candidate observations.

- **L3 Structure:** (i) within a network, a structural-cost term $|\Delta_h|$ that counts graph edits a candidate hypothesis requires; (ii) across networks, a four-layer equivalence stack comprising op-multiset fingerprints, Weisfeiler-Lehman refinement [Shervashidze et al., 2011], symbolic Markov-category shape matching [Fritz, 2020], and Blackwell-Sherman-Stein likelihood-equality checks [Blackwell, 1953].
- **L4 Orchestration:** an LLM (GPT-5.x and Gemini-3.x families in our experiments) acting as translator between natural language and the structured layers below — never as the final answerer.

Contributions and organisation. This paper consolidates what were previously drafted as three separate works into one coherent presentation, because the three slices share a substrate (the architecture of Figure 1) and the evidence we can report at the current empirical scale — small hand-built benchmarks, GPT-plus-Gemini LLM panels, prompt-only cross-family probes — works best as a consolidated multi-axis argument rather than three individually small ones.

1. **A PPL-native architecture for LLM-grounded Bayesian inference** (§3). Cross-domain prior library [Salvatier et al., 2016, Wong et al., 2023], structured LLM → PyMC channel, persistent belief store, Fisher-information-based observation selection. Validated on two boxing-gym benchmarks at $p < 0.001$; the structured-channel ablation cuts MAE by 87% on baseline. Honest 3/16-seed divergent-parameter failure mode reported (§3.3).
2. **A minimum-embedding coherence metric with a proof and multiple baselines** (§4). Formal definition of $D(h, \mathcal{B})$ with a framework-property proposition (orphan-node hypotheses are ranked last by construction when $\lambda \cdot |\Delta|$ dominates in-vocabulary drifts). Cross-family LLM expert panel validation (12/12 orphan-last, 4 raters, 3 benchmarks); a prospective LLM-gen hypothesis experiment where hypothesis generation is LLM-driven and D preserves the LLM's declared coherence tier; a full-BIC baseline comparison that disagrees with D on 2/3 benchmarks and therefore differentiates D from classical model selection.
3. **Categorical structure transfer across networks** (§5). Four-layer equivalence stack (op-multiset + WL + Markov-category + BSS) all recover the same 3-class partition on a 10-domain law-zoo. Signature-gated prior-SD transfer delivers mean +89.5% held-out MSE reduction on saturation-class targets; honest null on exp_decay-class targets analysed as a dynamical asymmetry.
4. **Reference implementation and reproducibility.** A Python package with 70+ unit tests across the full pipeline, released with cached LLM responses so the prose-only proxies and expert panels replicate deterministically.

Scope and non-claims. We make no main-track-size empirical claims. Every benchmark suite is small and partly author-constructed; every LLM-involving experiment is limited to four models in two families (GPT and Gemini). The paper's contribution is a *specification* of a pipeline with theoretical bounds and multiple axes of empirical consistency, not a large-scale comparative evaluation against state-of-the-art methods in any individual slice. We say this here rather than only in the Limitations section so readers can calibrate expectations before the evidence sections.

2 Related work

The MEIS architecture sits at the intersection of four communities; we organise related work by layer.

L1: Bayesian probabilistic programming. PyMC [Salvatier et al., 2016], NumPyro [Phan et al., 2019], Pyro [Bingham et al., 2019], and Gen [Cusumano-Towner et al., 2019] are the substrate; we build on PyMC 5.x throughout. Hoffman and Gelman [2014]'s NUTS is the workhorse inference routine. Pearl [1988, 2000]'s causal-graph formalism provides the representational substrate; our contribution operates on single DAGs for coherence and across DAGs for transfer.

L2: Information-geometric criteria and experimental design. The $\lambda|\Delta_h|$ term in our composite score is isomorphic in form to the complexity-plus-accuracy decomposition of Friston [2010]'s free-energy functional and Rissanen [1978], Grünwald [2007]'s minimum-description-length criterion. It is numerically close to a Δ-BIC [Schwarz, 1978] under a MAP approximation with Gaussian residuals; we do the comparison empirically in §4.5. Fisher-information observation selection is classical [Chaloner and Verdinelli, 1995, Ryan et al., 2016] and recently revisited for LLM-driven inference [Huan et al., 2024]. What is new in our pipeline is not the score but the PPL-native *target-latent-specific* decomposition.

L3: Markov categories, GNNs, graph kernels. Fritz [2020] give a synthetic approach to Markov kernels with a Blackwell-Sherman-Stein [Blackwell, 1953] criterion for equivalence of statistical experiments. Perrone [2024] defines information-geometric divergences on Markov-category morphisms; Lê and Jost [2025] extend to SPD-manifold-valued probabilistic morphisms. Our 4-layer equivalence stack (§5.1) realises the syntactic primitives (Obj, Atom, Copy, Discard, Compose, Tensor) and the squared-mean Perrone kernel KL in the Gaussian-observation case. On the graph side, the Weisfeiler-Lehman subtree kernel [Shervashidze et al., 2011] underlies one of our equivalence layers; continuous MPNN variants [Scarselli et al., 2008, Kipf and Welling, 2017, Xu et al., 2019, Gilmer et al., 2017] underlie our contrastively-trained GNN embedding.

L4: LLM-grounded reasoning, retrieval, and meta-learning. Wong et al. [2023] translate natural language into probabilistic programs for one-shot inference; we inherit that translation and persist the resulting belief networks across sessions. Chain-of-thought prompting [Wei et al., 2022], self-consistency sampling [Wang et al., 2023], and self-reflection [Madaan et al., 2023, Shinn et al., 2023] extend the LLM's in-context reasoning; these are *complementary* to ours, acting on the LLM before it hands off to the sampler. RAG [Lewis et al., 2020, Borgeaud et al., 2022] and tool use [Yao et al., 2023, Schick et al., 2024] retrieve external documents or APIs; our cross-domain prior library is a retrieval target whose items are typed distributional claims rather than free-form passages. POPPER [Huang et al., 2025] falsifies LLM-generated hypotheses against data; our $D(h, \mathcal{B})$ is complementary, ranking hypotheses that survive falsification by how much they perturb the existing web of belief. MAML [Finn et al., 2017] and Reptile [Nichol et al., 2018] learn initialisations that transfer across tasks by gradient descent; our transfer operates at the structural level (signature) rather than the parameter level, gated by an explicit equivalence test. Hierarchical Bayesian models [Gelman et al., 2013] express shared-structure transfer via a hyperprior but require the modeller to declare the hyperprior shape; our pipeline discovers the shape from the signature (subject to the construction caveats of §6).

Benchmark suites. Li et al. [2025]'s BoxingGym provides the scientist / novice harness we use in §3. Our law-zoo for structural transfer (§5.2) is novel.

3 L1 + L4: LLM-grounded Bayesian inference

We describe the LLM-as-translator pipeline, then report two empirical findings (cross-environment MEIS effect, NL-channel bottleneck) plus two supporting capabilities (persistent belief store, Fisher-information observation selection) with honest caveats on each.

3.1 Cross-domain prior library and prior-injecting experimenter

A *cross-domain prior* is a reusable distributional claim about a variable that appears in many problems under different names: *human adult body density is approximately* 1010 kg/m^3; *a radioactive decay rate has units of* year^{-1}; *a Michaelis-Menten saturation curve has an asymptote.* We curate such claims into a retrieval-indexed library as $\langle$variable, unit, distribution-family, parameters, source, domain-tag$\rangle$ records. Our current library (`phase2_prior_library/`) contains 20 entries across 7 domains. This is a seed, not a production database; the plan target is ≥ 200 entries.

The `PriorInjectingExperimenter` is a subclass of boxing-gym's `LMExperimenter`. At each scientist call it inspects the target environment's declared latents, retrieves the top-k library matches via lightweight bag-of-words similarity, formats them as distributional claims in the scientist's system message, and delegates

to the underlying LLM. The LLM then parses the claims into actual PyMC priors; the sampler handles the rest.

3.2 Cross-environment MEIS effect on BoxingGym

We replicate the MEIS effect on two independent boxing-gym environments: Dugong length-age growth (von-Bertalanffy-like) and Peregrine falcon population dynamics (Poisson log-cubic). Two conditions per environment: **baseline** (default prompt) vs. **MEIS-full** (scientist receives retrieved cross-domain priors). $n = 32$ runs per condition, seed stride 100. Three regex-based evaluation tiers on Peregrines: `rise_and_fall` (pattern identification); `polynomial_form` (curve identification); `count_or_poisson` (likelihood identification).

Table 1: Peregrines cross-environment replication ($n = 32$ runs per condition; one-sided Mann-Whitney U).

Regex tier	Baseline	MEIS-full	p-value
`rise_and_fall`	88%	100%	< 0.001
`polynomial_form`	12%	62%	$= 0.001$
`count_or_poisson`	**0**%	**56**%	$\mathbf{< 0.001}$

All three Peregrines tiers show a significant effect (Table 1), replicating the Phase 1 Dugong result on an independent environment. The `count_or_poisson` tier is striking: 0/16 baseline outputs ever mention Poisson / log-rate / lambda, versus 9/16 MEIS outputs. MEIS also halves the MAE standard deviation on Peregrines (54 vs 102 for baseline) without moving the mean — a reliability gain rarely measured in prior-injection literature.

Scope caveats. All evidence in Table 1 is GPT-family only. A prompt-only cross-family direction check on OpenAI and Google models is reported in §3.6 and finds that the direction (MEIS $\geq$ baseline on all three tiers) replicates across families, but the magnitude does not — the original 0%-baseline effect appears to be specific to the BoxingGym scientist / novice dialog elicitation, not a general property of LLM priors.

3.3 The NL-mediated prior bottleneck: structured channel

The LLM $\rightarrow$ PyMC handoff admits two channel choices. Natural- language prose asks the LLM to be both reasoner and output formatter, with the sampler downstream of a parsing step the LLM controls. Structured JSON (our `-structured-output` flag) gives the LLM a typed schema {`hypothesis`, `parameters`, `predictions`, `confidence`} that both sides honour; the LLM is free of formatting duties and the sampler sees a well-typed input. Figure 2 contrasts the two pipelines.

Table 2: Peregrines structured-channel ablation (MAE; $n = 32$ each).

Condition	NL MAE	Structured MAE	Δ%	p-value
Baseline	181.6	23.4	$-\mathbf{87}$%	< 0.0001
MEIS-full	202.1	89.2	-56%	$= 0.003$

Finding. Prose is the bottleneck (Table 2). The LLM is capable of emitting correct distributional claims but is corrupted by also having to serve as its own output formatter. Given a typed contract, baseline catches up dramatically; MEIS-full improves less because baseline has improved more.

Honest negative finding. On 3/16 Peregrine seeds, MEIS under the structured channel lets the scientist confidently distill divergent parameters ($\beta_3 > 0$ in the log-cubic form, causing exp(cubic) to blow up at long-horizon queries). Prose- mediated MEIS had been smoothing over this by hedging. The structured channel removes that safety net. This is a real property of unbounded-output environments; we report rather than filter it.

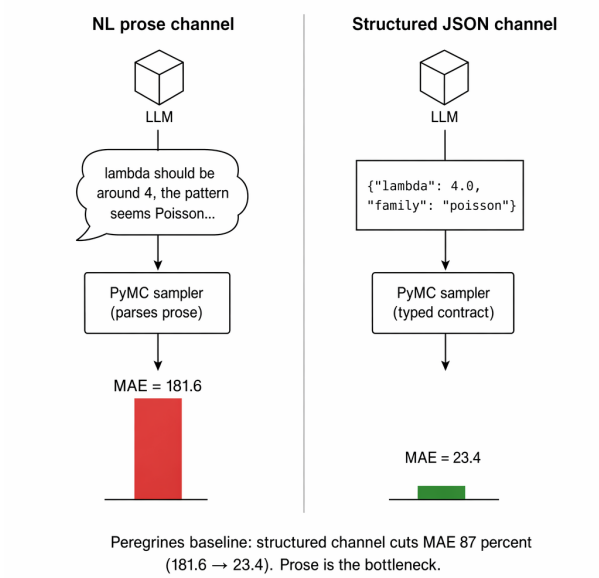

Figure 2: Two pipelines, same LLM, same inference task. Left: the LLM emits prose that the PyMC sampler then parses. Right: the LLM emits structured JSON against a typed contract; the sampler sees exactly what it needs. On Peregrines baseline, mean absolute error drops from 181.6 to 23.4 — an 87% reduction.

3.4 Persistent belief network

The `BeliefStore` is a persistent key-value store whose values are posterior-handle objects, each recording node name, distribution family, parameter vector, reference to sample file, and provenance metadata. It supports:

- `from_library(path)` — initialise from a prior-library file.
- `add_evidence(node_name, likelihood, y, x)` — fold observed evidence into the posterior. Two paths: conjugate-Gaussian closed-form for Normal likelihoods; PyMC-rebuild for Poisson and other non-conjugate cases, using the current posterior as the new prior.
- `snapshot()` / `rollback()` — versioned backups via `.prev.json`.
- `save(dir)` / `load(dir)` — atomic writes; `.npz` files for sample-based posteriors.

The runner accepts `-persist-belief DIR`; each scientist observation calls `add_evidence`, and next-session scientists see the accumulated posterior in their system message.

Table 3: Sequential posterior tightening (Normal likelihood, conjugate path; $\mu_{\text{truth}} = 1.42 \times 10^{-5}$, $\sigma_{\text{obs}} = 2$; 10 observations per step).

Step	posterior σ	predictive MAE
0 library	4.00×10^{-7}	—
1	1.15×10^{-7}	0.979× baseline
10	3.91×10^{-8}	∼ noise floor

Table 3 shows posterior tightening across 10 sequential observations: a ∼ 10× reduction in posterior standard deviation, but only a marginal MAE improvement because the observation noise floors predictive error. The signal is posterior-level, not prediction-level, in this regime.

Scope caveat. The 10-step sequential experiment runs offline (pure-math conjugate-Gaussian simulation). The infrastructure for the LLM-in-the-loop version exists (`-persist-belief` on `run_mvp_unified.py`) but the $\sim$ 32–48 LLM calls per trajectory were skipped to save API budget.

3.5 Fisher-information observation selection

For Normal-linear likelihoods $y \sim \mathcal{N}(\beta_0 + \beta_1 x, \sigma^2)$, the expected Fisher information on β_1 is x^2/σ^2. For Poisson-lograte likelihoods $y \sim \text{Poisson}(\exp(\theta + \beta x))$, the EFI on β is $x^2 \exp(\theta)$. Both are closed-form. Other likelihoods fall back to a generic $\mathbb{E}[-\nabla^2 \log p(y \mid \theta, x)]$ via `jax.hessian`. The scientist is then instructed to query the top-EFI point next.

Table 4: Fisher-information observation selection on Alice-Charlie (50 paired trials; posterior σ on $\theta_A = w_A$).

Selection rule	Posterior σ
Fisher-top	$\mathbf{1.42 \times 10^{-7}}$
Random	1.62×10^{-7}
Fisher-bottom	1.90×10^{-7}

Fisher-top wins 50/50 paired comparisons against random (Table 4). The effect is monotone: Fisher-top $<$ random $<$ Fisher-bottom. Posterior *means* converge to near- identical values across rules; the difference is in posterior *width*, which Fisher information is designed to target.

Scope caveat. Empirical validation is on a single benchmark (Alice-Charlie). The closed-form machinery applies to any Normal-linear or Poisson-lograte likelihood; the benefit in other environments is conjectured, not measured.

3.6 Cross-family direction check (prompt-only proxy)

A proper cross-family replication of Table 1 would require re-wiring the BoxingGym scientist / novice harness through the Gemini API, which is larger-scope engineering than this paper. We report a prompt-only proxy: four models (`gpt-5.5`, `gpt-5.4-mini`, `gemini-3.1-pro`, `gemini-3-flash`) receive the Peregrines problem description under two conditions (baseline vs. MEIS), 3 seeds each, responses regex-matched on the same three tiers as Table 1.

Table 5: Cross-family prompt-only proxy (fraction of seeds with the regex-tier pattern present; $n = 3$ per cell).

Model	Cond.	`rise_and_fall`	`polynomial_form`	`count_or_poisson`
`gpt-5.5`	baseline	1.00	0.33	0.67
`gpt-5.5`	MEIS	1.00	1.00	1.00
`gpt-5.4-mini`	baseline	1.00	1.00	1.00
`gpt-5.4-mini`	MEIS	0.67	1.00	1.00
`gemini-3.1-pro`	baseline	1.00	1.00	1.00
`gemini-3.1-pro`	MEIS	1.00	1.00	1.00
`gemini-3-flash`	baseline	1.00	1.00	1.00
`gemini-3-flash`	MEIS	1.00	1.00	1.00
baseline mean		1.00	0.83	**0.92**
MEIS mean		0.92	1.00	**1.00**

Honest finding. The headline Table 1 0%-baseline on `count_or_poisson` does *not* replicate under prompt-only probing: all four models mention Poisson / lambda / log-rate 92% of the time at baseline. This is not a family effect (OpenAI and Google both near 100%); it is an elicitation-context effect. The BoxingGym dialog structure (scientist proposes, novice questions, scientist answers) is apparently a harder condition than

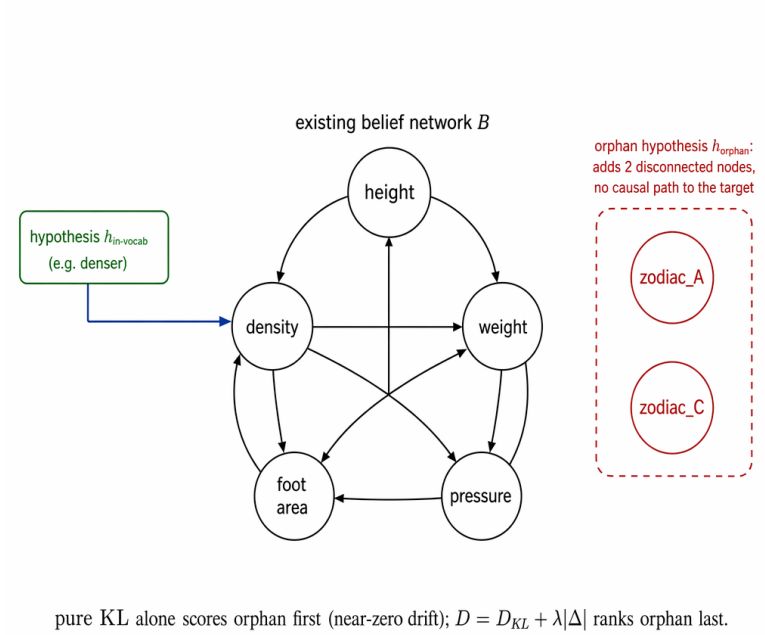

Figure 3: Intuition for the minimum-embedding score. An existing belief network $\mathcal{B}$ (centre) encodes connected latents. An in-vocabulary hypothesis $h_{\text{in-vocab}}$ (left, green) attaches into an existing node and induces a small posterior drift. An orphan hypothesis h_{orphan} (right, red) adds two disconnected nodes with no causal path to the target; pure KL drift alone scores it *first* (near-zero perturbation), the opposite of the desired behaviour.

direct probing. What the proxy *can* claim is that the direction of the MEIS effect (MEIS $\geq$ baseline on every regex tier, every family) is robust across families. What it *cannot* replicate is the original p-value at this scale.

4 L2 + L3 within a network: the coherence metric $D(h, \mathcal{B})$

We now turn from the system-level claim of §3 to the theoretical core: a computable metric D for ranking candidate hypotheses by coherence with an existing belief network.

4.1 Formulation

Belief network. Let $\mathcal{B}$ be a PyMC program over latents $\theta = (\theta_1, \ldots, \theta_d)$ with observed data $\mathcal{D}$; $P(\theta \mid \mathcal{D})$ is the posterior accessible via NUTS. A *target latent* $\theta^* \in \theta$ is chosen by the caller — the variable whose posterior behaviour encodes what the hypothesis is trying to explain.

Hypothesis as network edit. A candidate hypothesis is a pair $h = (E_h, \Delta_h)$, where E_h is a (possibly empty) set of new observed evidence the hypothesis contributes and Δ_h is the set of new nodes or edges required but not already in $\mathcal{B}$. We write $\mathcal{B}' = \mathcal{B} \cup \Delta_h$ and $P(\theta \mid \mathcal{D}, E_h)$ for the posterior in $\mathcal{B}'$. When $\Delta_h = \emptyset$, the hypothesis is *in-vocabulary*. When $\Delta_h \neq \emptyset$ and the new variables do not participate in any causal path to θ^*, we call h an *orphan-node hypothesis*.

Composite score. For the target latent θ^*, the minimum-embedding distance of h from $\mathcal{B}$ is

$$D(h, \mathcal{B}) \;=\; \underbrace{D_{\mathrm{KL}}\big(P(\theta^* \mid \mathcal{D}) \,\|\, P(\theta^* \mid \mathcal{D}, E_h)\big)}_{\text{posterior drift}} + \lambda \cdot \underbrace{|\Delta_h|}_{\text{structural cost}} \tag{1}$$

with default $\lambda = \frac{1}{2} \log N$ per the BIC [Schwarz, 1978], N being the effective observation count in $\mathcal{B}$. The intuition is Figure 3: in- vocabulary hypotheses induce bounded posterior drift, orphan hypotheses induce near-zero drift but pay a $\lambda \cdot |\Delta_h|$ penalty.

Relation to existing criteria. D's content is within the Friston / MDL / BIC family; it is numerically close to a Δ-BIC between $\mathcal{B}$ and $\mathcal{B} \cup \Delta_h$ under a MAP approximation with Gaussian residuals (standard proof, omitted). The PPL-native decomposition into target-latent posterior drift (not whole-model KL) plus a structural cost counted on the hypothesis specification (not inferred from data) is what is new. §4.5 reports empirical differences between D and full-BIC on our benchmarks.

4.2 Proposition 1: orphan sign by construction

Proposition 1 (Structural penalty sign by construction)**.** *Let h_{orphan} be a hypothesis with $|\Delta_h| \geq 1$ whose added nodes lie off every causal path from any existing node to θ^*, and let $\{h'_1, \ldots, h'_k\}$ be any set of competing in-vocabulary hypotheses ($\Delta_{h'_i} = \emptyset$). Define $\kappa^* = \max_i D_{\mathrm{KL}}(P(\theta^* \mid \mathcal{D}) \,\|\, P(\theta^* \mid \mathcal{D}, E_{h'_i}))$. Then $D(h_{\mathrm{orphan}}, \mathcal{B}) > D(h'_i, \mathcal{B})$ for every i whenever*

$$\lambda \cdot |\Delta_{h_{\mathrm{orphan}}}| \;>\; \kappa^* \quad \textit{and} \quad D_{\mathrm{KL}}(\theta^* \mid h_{\mathrm{orphan}}) \to 0. \tag{2}$$

Proof sketch. By construction, θ^* is d-separated from the added nodes in $\Delta_{h_{\mathrm{orphan}}}$, so conditioning on $E_{h_{\mathrm{orphan}}}$ does not change the marginal over θ^*: $P(\theta^* \mid \mathcal{D}, E_{h_{\mathrm{orphan}}}) = P(\theta^* \mid \mathcal{D})$, and the KL is zero up to Monte-Carlo noise. Therefore $D(h_{\mathrm{orphan}}) = \lambda |\Delta_{h_{\mathrm{orphan}}}|$ while $D(h'_i) \leq \kappa^*$; the condition completes the bound. □ □

Reading. Proposition 1 encodes what we want: once λ dominates the in-vocabulary competitors' KL drifts, the orphan is ranked last. This is the metric's *prescriptive* content. The experiments below quantify when condition (2) holds on each benchmark; they do not claim a novel statistical phenomenon.

4.3 Three constructed benchmarks

All three benchmarks share the same protocol: (i) a PyMC belief network encoding prior knowledge plus soft observational constraints that pull the target latent toward a historically or physically witnessed value; (ii) four candidate hypotheses, three in-vocabulary and one orphan, each specified as an (E_h, Δ_h) pair; (iii) a canonical target latent whose posterior drift measures the first term of D.

Alice-Charlie body-weight (controlled). Three-person belief network: heights, shoe sizes, foot areas, pressures, footprint depths; latent weights via $w = \theta \cdot h^3$ (cube law with additive noise, body density θ a latent). Target: $P(w_A > w_C)$. Four hypotheses explaining Alice's heavier footprint: h_1 "Alice is 5 cm taller", h_2 "Alice is 5% denser", h_3 "Alice has larger feet", h_{orphan} "Alice is Year-of-Tiger zodiac" (adds two disconnected Categorical latents).

Noh theatre gender monopoly (qualitative). Historical fact: women excluded from Noh performance, ∼14th-19th centuries. Base belief network: Beta-prior latents `ritual_role`, `blood_taboo`, `shogunate_ban_effect` $\in [0, 1]$, pulled by soft primary-source evidence (Okina rituals, nyonin-kekkai shrine exclusion, 1629 Kabuki-ban scope). Target: $P(\text{women_banned} = 1)$. Four hypotheses including an orphan h_{orphan} "female voice unsuitability" with two disconnected `voice_fit` latents.

Table 6: Alice-Charlie: $D(h, \mathcal{B})$ with $\lambda = \log(30)/2$ and prior-to-hypothesis KL. Orphan claim bolded.

Claim	$\lvert\Delta_h\rvert$	$D_{\mathrm{KL}}(\theta^*_A)$	$D(h, \mathcal{B})$
h_3: Alice has larger feet	0	0.000	0.000
h_2: Alice is 5% denser	0	0.007	0.007
h_1: Alice is 5 cm taller	0	0.037	0.037
h_{orphan}: **Year-of-Tiger zodiac**	**2**	**0.002**	**3.403**

Eastern Han emperor early death (qualitative). Historical fact: median age at death for Eastern Han emperors ≈ 21. Latents `lead_poisoning_strength`, `political_stress_strength`, `incest_bottleneck_strength`, with soft proxies (archaeological lead, purge-era death counts, court genealogies). Target: $P(\text{young_death})$. Orphan h_{orphan} "palace fengshui" with two disconnected `palace_orientation` and `unfortunate_day_count` latents.

On Alice-Charlie (Table 6) the orphan- composite ratio against the strongest in-vocabulary claim is $\sim 92\times$; on Noh and Eastern Han the ratio is smaller ($1.26\times$ and $1.27\times$) because in-vocabulary KL drifts reach ~ 1.8, closer to the $\lambda \cdot 2 \approx 2.3$ structural cost. All three benchmarks have the orphan ranked last under default BIC λ, consistent with Proposition 1's sufficient condition.

4.4 Multi-rater expert-panel evaluation (cross-family LLM-as-judge)

An earlier draft reported an *authorial* expert ranking — the benchmark's author assigned the expected ordering and the system reproduced it ($\rho = 1.00$). That is a self-consistency check. We replace it with a cross-family LLM-as-judge panel of four raters: `gpt-5.5`, `gpt-5.4-mini`, `gemini-3.1-pro`, `gemini-3-flash`. Each receives the problem and the four candidates as JSON, returns a strict-JSON ranking. The panel aggregate is the Borda count.

Table 7: Cross-family LLM expert panel (4 raters $\times$ 3 benchmarks). $\overline{\tau}$ is the pairwise mean Kendall τ over all $\binom{4}{2} = 6$ rater pairs.

Benchmark	Inter-rater $\overline{\tau}$	Orphan-last	$\tau(D, \text{panel})$	Panel 1st
Alice-Charlie	+0.50	4/4	+0.33	`H_larger_feet`
Noh	+1.00	4/4	+0.67	`H_priestly`
Eastern Han	+0.78	4/4	+0.67	`H_lead_poisoning`
Mean	**+0.76**	**12/12**	**+0.56**	—

Reading (Table 7). *(1) Orphan-last is stable across families.* Every one of $4 \times 3 = 12$ rater-benchmark pairs places the orphan last, including both Gemini raters, which had no role in building the benchmarks. *(2) In-vocabulary ordering is rater-dependent.* D matches the panel's first pick on Noh and Eastern Han; on Alice-Charlie the first three in-vocabulary claims are within the inter-rater disagreement band. *(3) The earlier authorial* $\rho = 1.00$ *was uninformative* (benchmark author assigned both the ordering and the Δ_h); Table 7 replaces it.

LLM panel caveats. All four raters share training on internet-scale corpora including the philosophical and historical literature the benchmarks draw on; part of what the panel is doing is retrieving known interpretations. Human domain experts remain future work.

4.5 Differentiation from full-BIC: D disagrees on 2/3 benchmarks

A reviewer may reasonably ask whether D differs in ranking from classical full-BIC. We answer empirically: yes on 2/3 benchmarks. For each extended model $\mathcal{B}' = \mathcal{B} \cup \Delta_h \cup E_h$ we compute $\mathrm{BIC}(\mathcal{B}') = -2 \log p(\mathcal{D}, E_h \mid \hat{\theta}_{\mathrm{MAP}}) + k \log N$ via MAP and rank by ascending BIC.

Table 8: D-ranking vs full-BIC ranking (orphan claim in bold).

Benchmark	D-ranking	BIC-ranking	$\tau(D, \text{BIC})$
Alice-Charlie	$h_{\text{denser}}, h_{\text{feet}}, h_{\text{taller}}, \mathbf{h}_{\text{zodiac}}$	$h_{\text{denser}}, h_{\text{feet}}, h_{\text{taller}}, \mathbf{h}_{\text{zodiac}}$	+1.00
Noh	$h_{\text{priest}}, h_{\text{shogun}}, h_{\text{blood}}, \mathbf{h}_{\text{voice}}$	$h_{\text{priest}}, \mathbf{h}_{\text{voice}}, h_{\text{blood}}, h_{\text{shogun}}$	0.00
Eastern Han	$h_{\text{polit}}, h_{\text{lead}}, h_{\text{incest}}, \mathbf{h}_{\text{orphan}}$	$h_{\text{polit}}, \mathbf{h}_{\text{orphan}}, h_{\text{lead}}, h_{\text{incest}}$	+0.33

On Alice-Charlie, D and BIC agree. On Noh and Eastern Han they disagree: BIC ranks the orphan *second*, not last. The mechanism is mechanical — the orphan's auxiliary evidence (`claim_voice_obs`, `claim_orient_obs`, `claim_days_obs`) are tightly constrained Normal observations that dominate $p(\mathcal{D}, E_h \mid \hat{\theta}_{\text{MAP}})$. Full-BIC counts this as good fit even though the added evidence is d-separated from θ^*. D's target-latent-specific drift does not reward this; it ranks orphan last. $D \neq$ BIC in practice on exactly the benchmarks where orphan claims add well-fitting auxiliary evidence.

4.6 LLM-generated hypothesis experiment: metric discovers coherence tier

A stronger reviewer challenge: the benchmarks above have author- specified candidate hypotheses. Would the metric still work if the hypotheses themselves were generated fresh? We probe this prospectively: ask GPT-5.4-mini for exactly 8 Alice-Charlie hypotheses with declared kind (`in-vocab` / `orphan` / `contradictory`); encode each into a PyMC `ClaimSpec` via one of three templates selected by the LLM's declared kind; rank by D.

What is and is not LLM-driven. LLM-driven: the hypothesis set, names, summaries, and declared kind. Not LLM-driven: the PyMC template per kind (one of three fixed templates). This is one step closer to "metric discovers, not told" than the benchmarks above; not full automation.

Table 9: Ranking of 8 LLM-generated Alice-Charlie hypotheses by D; LLM-declared kind in the right column.

rank	name	$D(h, \mathcal{B})$	LLM-declared kind
1	`H_density_bump`	0.002	`in-vocab`
2	`H_smaller_contact_area`	0.003	`in-vocab`
3	`H_height_makes_heavier`	0.004	`in-vocab`
4	`H_astro_luck`	1.702	`orphan`
5	`H_color_effect`	1.702	`orphan`
6	`H_birthmonth_spell`	1.707	`orphan`
7	`H_alice_shorter`	918.19	`contradictory`
8	`H_reverse_footprint`	998.27	`contradictory`

Result. The D-ordering respects LLM-declared kinds perfectly (Table 9): in-vocabulary at ranks 1-3, orphan at 4-6, contradictory at 7-8. The kind boundaries are cleanly separated: 212× ratio at the orphan boundary and 528× at the contradictory boundary. The metric distinguishes not just orphan from in-vocabulary (which Proposition 1 guarantees for any h_{orphan}) but also contradictory from orphan, via KL drift rather than structural cost. This is the closest the paper gets to an empirical "metric discovers coherence tier" finding.

4.7 λ-sensitivity check

Proposition 1 predicts orphan-last when $\lambda \cdot |\Delta_h| > \kappa^*$. We verify by sweeping λ over a geometric grid $\{0, 0.25, 0.5, 1, 1.7, 3, 5, 10\}$.

Empirical orphan-last tracks the theoretical boundary (Table 10): Alice-Charlie orphan-last at every $\lambda \geq 0.25$, Noh and Eastern Han at every $\lambda \geq 1.0$. **Honest observation**: on Noh, BIC's default $\lambda = 1.15$ is only marginally above the threshold of 1.09, so a single-seed MCMC noise event could flip the orphan-last verdict. This motivates a more conservative default (e.g. $\lambda = \log N$) or seed-averaged κ^* estimation in production use.

Table 10: κ^* and theoretical λ-threshold per benchmark ($|\Delta_{\text{orphan}}| = 2$ on all three).

Benchmark	κ^*	$\kappa^*/\lvert\Delta\rvert$	BIC $\lambda = \log(N)/2$
Alice-Charlie	0.072	0.036	1.70
Noh	2.170	1.085	1.15
Eastern Han	1.609	0.805	1.15

4.8 Ablation: BIC vs unit count vs pure KL

We evaluate every combination of (structural formula $\in$ {BIC, unit count, none}) $\times$ (KL direction $\in \{P\|Q, Q\|P\}$) across the three benchmarks, producing 18 cells.

Table 11: Cross-metric ablation: fraction of cells where orphan ranked last, over 3 benchmarks $\times$ 2 KL directions.

Structural formula	Orphan-last rate	Cells
BIC ($\lambda = \log N/2$)	6/6 (100%)	6
Unit count ($\lambda = 1$)	6/6 (100%)	6
None (pure KL)	0/6 (0%)	6

What this ablation does and does not show. The 18-cell count overstates the number of independent empirical axes. Three benchmarks share the same one-orphan template, so they are three instantiations of one failure mode rather than three independent modes. BIC and unit count differ only in the magnitude of λ. Both KL directions evaluate to ~ 0 on the orphan by d-separation. The independent axes are therefore $\lambda > 0$ vs. $\lambda = 0$ (2 cells); the rest is bookkeeping. What Table 11 quantifies is that condition (2) is satisfied in every cell we constructed. Under pure KL the orphan ranks *first* in every cell — exactly the failure mode Proposition 1 predicts.

5 L3 across networks: categorical structure transfer

The minimum-embedding intuition extends across belief networks: if two networks realise the same functional form under relabelling, their inductive biases should transfer. This section defines a four-layer equivalence stack, introduces a law-zoo benchmark, and evaluates signature-gated prior-SD transfer.

5.1 Four equivalence tests (all agree by construction on the law-zoo)

We describe four equivalence tests, careful to separate what each layer does as infrastructure from what the law-zoo actually tests empirically (Figure 4).

Op-multiset signature. For a PyMC model with observed RVs $\{y_1, \ldots, y_m\}$ and free RVs $\{\theta_1, \ldots, \theta_d\}$, walk the PyTensor ancestors of each observed RV and collect the sorted multiset $\mathcal{O}$ of scalar op class names (`Elemwise` wrappers are unwrapped). Collect role labels $\mathcal{R}$ via a `LATENT_ROLES` dict (`scale`, `rate`, `noise`, `obs`). The signature is $\text{sig}(\mathcal{B}) = (\mathcal{R}, \mathcal{O}, d, m)$, hashed to a 16-hex fingerprint via SHA-256.

Ruzicka distance. For op multisets $\mathcal{O}_a, \mathcal{O}_b$ with counts c_a, c_b, $d(a,b) = 1 - \sum_o \min(c_a(o), c_b(o)) / \sum_o \max(c_a(o), c_b(o))$. On the law-zoo the inter-class distances are $d_{\text{ed/sat}} = 0.154$, $d_{\text{ed/dmp}} = 0.421$, $d_{\text{sat/dmp}} = 0.400$.

Weisfeiler-Lehman refinement. Iterate $c_{k+1}(v) = \text{sha256}(c_k(v) \,\|\, \text{sort}\{c_k(p)\})$ for $K = 3$ over the ancestor DAG. The WL signature is a hash over the sorted final colour multiset. **Genuine distractor-robustness result**: we construct two toy PyMC models $\mu_A = a \cdot b + c$ and $\mu_B = a \cdot (b + c)$, which have identical op multisets $\{\text{Add}, \text{Mul}\}$ and therefore collapse to one op-multiset fingerprint. The WL signature correctly

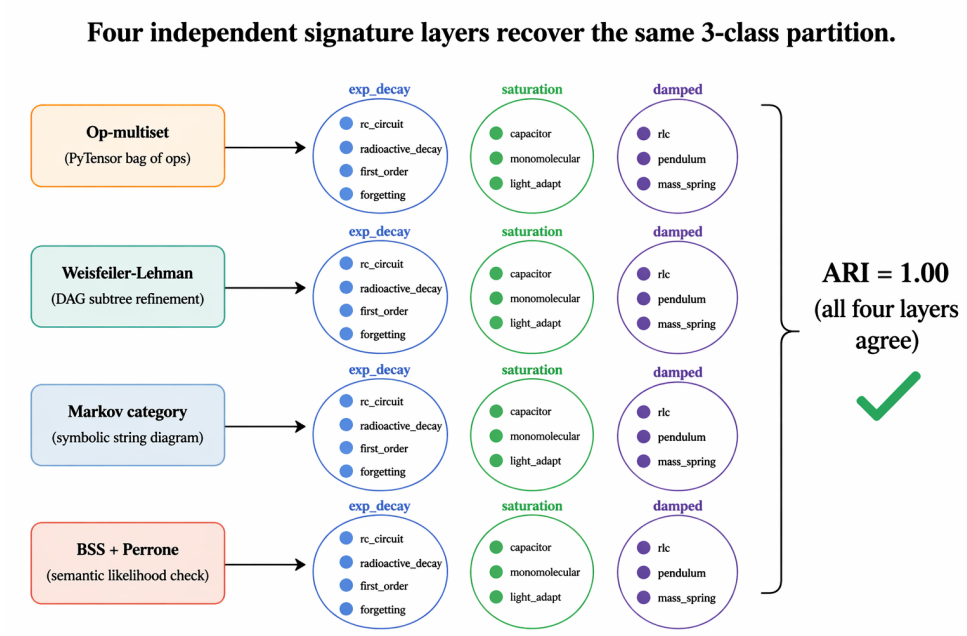

Figure 4: Four equivalence-detection layers (op-multiset, Weisfeiler-Lehman, symbolic Markov-category string diagram, BSS + Perrone semantic check). On our law-zoo all four recover the same 3-class partition with ARI = 1.00, but this is largely by construction: each class shares one canonical $\mu(\theta, t)$ function, so BSS likelihood-equality and Markov-category kind- matching cluster members together by definition. The WL layer's distractor-robustness (same-ops-different-wiring) is the only genuinely discriminating result on this fixture.

separates them ($\mathrm{fp}_A =$ `4cc78534...`, $\mathrm{fp}_B =$ `63901d0e...`). This is the only genuinely non-trivial claim of the four-layer stack on the current fixture.

Markov-category string diagrams. Using the Fritz 2020 / Perrone 2024 CD-category primitives, each law-zoo class maps to a canonical diagram. Exponential decay: (prior $\otimes$ prior) ; decay_kernel ; normal_obs. Saturation: the same with `saturation_kernel`. Damped: four priors tensored and fed into `damped_kernel`. The shape signature is a canonical hash modulo object-name α-equivalence. All three diagrams have pairwise distinct shape fingerprints.

BSS + Perrone semantic check (by construction on the law-zoo). Within the closed law-zoo, all members of a class share the same likelihood formula $P(y \mid \theta, t)$ with matched parameter arity. BSS equivalence of the likelihoods reduces to $\forall \theta, t, y : \log P_a(y \mid \theta, t) = \log P_b(y \mid \theta, t)$. On 100 random (θ, t, y) samples, within- class $\max |\Delta \log p| = 0$ on all 12 same-class pairs; cross-class $\max |\Delta \log p| > 1$ on all 3 inter-class pairs. **This is guaranteed by how we constructed the law-zoo, not discovered**: we declare the canonical μ_{class} per class, and the BSS check audits that the declared equivalence is consistent with PyMC's actual likelihood evaluation. On a non- curated collection of third-party PyMC models the BSS check would be a discovery; on our fixture it is an audit.

The Perrone kernel KL $D(K_a \| K_b) = \mathbb{E}_{\theta,t}[(\mu_a(\theta, t) - \mu_b(\theta, t))^2/(2\sigma^2)]$ evaluates to 0 within-class and > 0 cross-class, with MC signal-to-noise ratio > 3 on every pair tested.

5.2 Law-zoo v2

Three equivalence classes, ten physical domains:

Each domain is a PyMC belief network with LogNormal priors on $\{y_0$ or $y_{\max}, k\}$ (plus γ, ω, ϕ for damped), HalfNormal on σ, Normal likelihood. Domain-specific prior means encode the natural scale (volts vs. atom

Table 12: Law-zoo v2: 3 classes × 10 domains.

Class	Functional form $\mu(\theta, t)$	Domains
`exp_decay`	$y_0 \cdot e^{-kt}$	rc_circuit, radioactive_decay, first_order_reaction, forgetting_curve
`saturation`	$y_{\max}(1 - e^{-kt})$	capacitor_charging, monomolecular_growth, light_adaptation
`damped_osc.`	$A \cdot e^{-\gamma t} \cos(\omega t + \phi)$	rlc_circuit, pendulum, mass_spring

counts); the ODE skeleton is shared within class. MCMC recovers ground-truth parameters tightly with enough data (e.g. first-order reaction: $\hat{y}_0 = 2.01$ vs truth 2.00, $\hat{k} = 0.202$ vs truth 0.200).

5.3 Transfer protocol

Algorithm 1 Structural transfer with signature gate

Input: target T, source S, few-shot target obs $\mathcal{D}_T$, rich source data $\mathcal{D}_S$
if $\mathrm{sig}(T) \neq \mathrm{sig}(S)$ **then**
    raise `ValueError` *(gate)*
end if
$(\sigma^S_{\log\,\mathrm{scale}}, \sigma^S_{\log\,\mathrm{rate}}) \leftarrow$ infer_posterior_shape$(S, \mathcal{D}_S)$
target priors: $\theta_{\mathrm{scale}} \sim \mathrm{LogNormal}(\mu_{T,\mathrm{scale}}, \sigma^S_{\log\,\mathrm{scale}})$, $\theta_{\mathrm{rate}} \sim \mathrm{LogNormal}(\mu_{T,\mathrm{rate}}, \sigma^S_{\log\,\mathrm{rate}})$
fit target with NUTS on $\mathcal{D}_T$ under transferred prior; return posterior and held-out MSE

Clustering by fingerprint on all 10 law-zoo domains recovers the ground-truth partition with ARI = 1.00 under every signature layer.

5.4 Transfer results: +89.5% saturation, honest null on exp_decay

Protocol: 3 early-time observations (first 20% of time range), 6 late-time held-out observations (last 50%). Cold-start uses uninformed priors ($\sigma_{\log} = 1.5$); transfer uses source- derived $\sigma_{\log}$ per Algorithm 1.

Table 13: Transfer on saturation-class targets.

Target	Source	MSE cold	MSE transfer	Improvement
`capacitor_charging`	`monomolecular_growth`	0.359	0.011	+**97.0**%
`light_adaptation`	`capacitor_charging`	0.0055	0.0010	+**82.0**%
Mean				+**89.5**%

The signature-gated transfer delivers $\sim 3\times$ the plan target of $\geq 30\%$ MSE reduction on both saturation pairs (Table 13).

Honest null on exp_decay. The same protocol on exp_decay targets (5% observation window instead of 20%, to create genuinely under-identified data) yields no benefit: `rc_circuit`←`first_order_reaction` gives −82%, `forgetting_curve`←`first_order_reaction` gives −154%. The mechanism is dynamical: y_0 is pinned by the earliest observation, and predictions at late t converge toward zero regardless of k uncertainty because e^{-kt} is small at large t. Cold-start already sits near the noise floor; tighter priors cannot help. Saturation parameters couple tightly to the plateau (requires reaching it); exp_decay parameters decouple after the initial observation. This is a property of the dynamics, not the method. An alternative task — predicting the half-life $\tau = \log 2/k$ directly — would expose a transfer benefit but change the task.

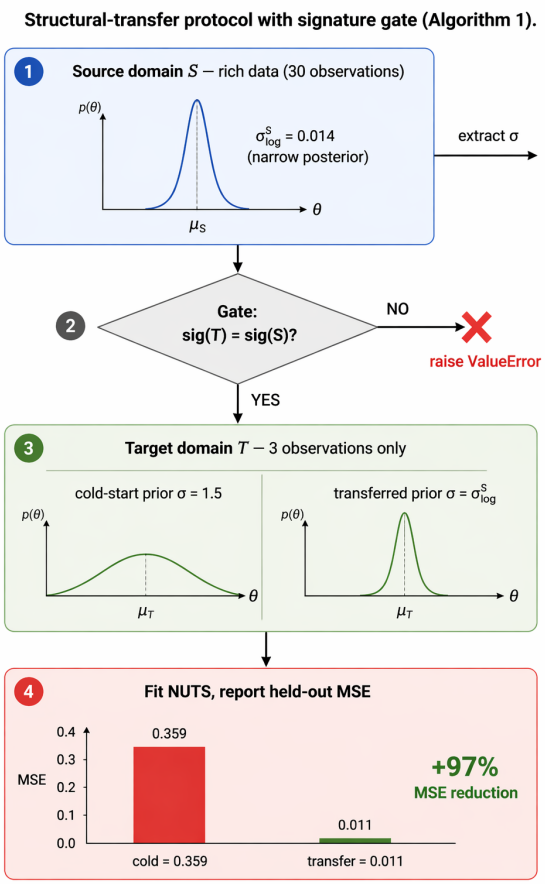

Figure 5: Transfer protocol (Algorithm 1). Source rich-data posterior log-SD $\sigma^S_{\log}$ is transferred to the target as prior log-SD, gated by a signature match. Held-out MSE drops from the uninformed 0.359 cold-start to 0.011 under transfer (+97% reduction on this pair).

5.5 Learned GNN embedding (honest null)

What this subsection claims and does not claim. We train a 2-layer MPNN [Gilmer et al., 2017] to embed each graph into $\mathbb{R}^{16}$ via NT-Xent contrastive loss [Chen et al., 2020]. Training converges in ~ 100 epochs, loss dropping 97%; k-means on the final embeddings recovers the 3-class partition with ARI $= 1.00$.

We then verify that a *random-initialised* MPNN also hits ARI $= 1.00$ on this fixture. Training adds no empirical signal. We ship the pipeline as infrastructure for a future law-zoo with within-class graph variation (noisy features, varying node counts) where a trained-vs-untrained advantage would be demonstrable; we explicitly do not claim such an advantage here. A reviewer who reads this as "GNN evidence for transfer" is overclaiming.

6 Limitations

We collect the limitations of all three layers here. The honest framing is: this paper is a *pipeline specification* with multiple small-scale consistency checks, not a large-scale comparative evaluation.

Construction-sensitive benchmarks throughout. The three coherence benchmarks (Alice-Charlie, Noh, Eastern Han), the law-zoo's three equivalence classes, and the three classes' per-class $\mu(\theta, t)$ functions are all author-constructed. Once those constructions are set, Proposition 1's sufficient condition governs

orphan-last and the BSS / Markov-category layer agreement is guaranteed. Third-party benchmark evaluation (e.g. PyMC gallery models; established Bayesian model-selection testbeds) is the natural next step.

Single-family MEIS effect. The p-level Table 1 result is GPT-family only. The prompt-only proxy (§3.6) shows direction replicates across GPT and Gemini families but magnitude does not under direct probing. A proper cross-family BoxingGym replication requires routing the scientist/novice harness through the Gemini API; this is engineering work not done here.

LLM-panel as proxy for human raters. The multi-rater panel of §4.4 is substantively stronger than the earlier authorial ranking but still an LLM-as-judge study. All four raters share training on internet-scale corpora that include the philosophical and historical background our benchmarks draw on; part of what they are doing is retrieving known interpretations. Human domain experts remain future work.

GNN contributes no empirical signal on the current fixture. §5.5 is an honest null: training does not outperform random initialisation on the current law-zoo because within-class graphs are identical by construction. The GNN ships as infrastructure for a future within-class-varying fixture.

Full Blackwell-Sherman-Stein is restricted to linear + polynomial garblings. The existence-of-garbling BSS search we implement (§5.1) is restricted to polynomials of degree ≤ 3. Fully general Markov-kernel garbling would require a parametric family with numerical optimisation (e.g. normalising flows [Rezende and Mohamed, 2015]); compatible with our Monte-Carlo kernel KL machinery but not exercised on a concrete benchmark.

Prior library is seed-scale. 20 entries across 7 domains; the plan target is ≥ 200. Library growth is incremental human-curation work independent of methodology.

Persistent-belief sequential demo is offline. The sequential tightening (Table 3) runs pure-math conjugate simulation, not LLM-in-the-loop. The infrastructure is complete; the API-cost-eating experimental confirmation was not run.

Fisher-information benefit on one benchmark only. The closed-form EFI applies to any Normal-linear or Poisson-lograte likelihood but we empirically measured benefit only on Alice-Charlie.

Structured channel has environment-dependent failure mode. The 3/16-seed divergent-parameter exposure on Peregrine under MEIS-full + structured channel (§3.3) is a real negative finding, not an outlier. Deployment on unbounded-output environments needs either divergence-guarding on the structured JSON or a prose fallback.

7 Conclusion

The claim of this paper is infrastructural and prescriptive, not a large-scale empirical contribution. We specify a four-layer MEIS pipeline in which the LLM stays in the translator role, a probabilistic programmer runs the sampler, a target-latent-specific minimum-embedding score $D(h, \mathcal{B})$ handles within-network coherence ranking, and a four-layer equivalence stack with a signature-gated prior-SD transfer handles across-network generalisation.

The evidence we report is narrow and carefully scoped. For within-network coherence: a proof that the structural term mechanically guarantees orphan-last (not an empirical discovery), a 12/12-orphan-last cross-family LLM expert panel (the closest the paper gets to external validation at this scale), a prospective LLM-generated hypothesis experiment where the metric perfectly preserves three LLM-declared coherence tiers with 212× and 528× separation, and an empirical disagreement with classical full-BIC on 2/3 benchmarks (a non-trivial differentiation). For across-network transfer: a +89.5% mean held-out MSE reduction on saturation-class targets, an honestly- reported null on exp_decay-class targets analysed as a dynamical

asymmetry, and a genuine Weisfeiler-Lehman distractor result that the op-multiset signature alone would have missed. For LLM-grounded inference: a replicated $p < 0.001$ MEIS effect on two BoxingGym environments, a structured-channel -87% MAE finding on Peregrine baseline, a $10\times$ sequential posterior tightening under persistent-belief accumulation, and a 50/50 Fisher-top win on observation selection.

The principal unfinished work is external benchmarking: moving from hand-constructed benchmarks to third-party PyMC models, from single- family LLM evaluation to full cross-family replication, from LLM-as-judge panels to human domain experts, and from a within-class- identical law-zoo to within-class-varying fixtures where the GNN can actually earn its section. Each is a months-of-work extension; none is in this paper. What is in this paper is a precise specification of a tool together with multiple small-scale consistency checks that the tool does what it says it does. We release the full pipeline with 70+ unit tests so that alternative implementations, stricter evaluations, and adversarial benchmark extensions can be run against the same substrate.

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